FOP Mini Project

Restaurant Bill Generator using C

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Objective

THE PRIMARY OBJECTIVE OF THIS PROJECT IS TO DESIGN AND DEVELOP A MENU-DRIVEN RESTAURANT BILLING SYSTEM USING THE C PROGRAMMING LANGUAGE THAT AUTOMATES THE PROCESS OF FOOD ORDERING AND BILL GENERATION. THIS SYSTEM AIMS TO SIMPLIFY THE MANUAL BILLING PROCESS BY ACCURATELY CALCULATING THE TOTAL AMOUNT BASED ON SELECTED ITEMS, QUANTITY, DISCOUNTS, SERVICE CHARGES, AND GST.

FEATURES OF THE PROGRAM

1. DO–WHILE LOOP (MENU REPETITION):

USED TO REPEAT THE ORDERING PROCESS UNTIL THE USER DECIDES TO STOP.
ENSURES THE MENU SHOWS AT LEAST ONCE.
PERFECT FOR "ORDER AGAIN?" TYPE PROGRAMS.
CONCEPT USED: LOOP CONTROL + USER-DRIVEN ITERATION

2.SWITCH–CASE STATEMENT (MENU SELECTION):

HANDLES MULTIPLE MENU OPTIONS CLEANLY.
EACH CASE UPDATES BILL + ITEM QUANTITY.
MUCH CLEANER THAN USING MANY IF–ELSE.
CONCEPT USED: MULTI-WAY DECISION MAKING

3. INPUT / OUTPUT FUNCTIONS:

SCANF() → TAKES USER INPUT (CHOICE, QUANTITY, COUPON)
PRINTF() → DISPLAYS MENU, BILL, RECEIPT
CONCEPT USED: CONSOLE I/O IN C

4.ARITHMETIC OPERATIONS:

MULTIPLICATION → PRICE × QUANTITY
PERCENTAGE CALCULATIONS → DISCOUNT, GST, SERVICE CHARGE
CONCEPT USED: NUMERIC COMPUTATION

5.VARIABLES FOR STATE TRACKING:

NTEGER VARIABLES → BILL, QUANTITIES, CHOICES
FLOAT VARIABLES → DISCOUNTS, GST, FINAL AMOUNT
SEPARATE ITEM COUNTERS (I1–I10) → TRACK ORDERED ITEMS
CONCEPT USED: DATA STORAGE & STATE MANAGEMENT

6.CONDITIONAL STATEMENTS:

IF CONDITIONS USED WHEN PRINTING BILL DETAILS:
IF(I1 > 0)
PRINTS ONLY ORDERED ITEMS.
CONCEPT USED: CONDITIONAL EXECUTION

WORKING OF THE PROGRAM

THE PROGRAM STARTS BY DISPLAYING A MENU INSIDE A DO-WHILE LOOP, ALLOWING THE USER TO SELECT ITEMS AND ENTER QUANTITIES MULTIPLE TIMES. A SWITCH-CASE STRUCTURE CALCULATES THE COST AND UPDATES THE TOTAL BILL AND ITEM COUNTERS AFTER EACH SELECTION.

ONCE ORDERING IS FINISHED, THE PROGRAM SHOWS THE TOTAL BILL AND ASKS THE USER TO CHOOSE A COUPON. BASED ON THE SELECTION, IT APPLIES EITHER A PERCENTAGE DISCOUNT OR A FLAT REDUCTION TO THE BILL.
AFTER DISCOUNT CALCULATION, THE PROGRAM ADDS A 10% SERVICE CHARGE AND 5% GST TO COMPUTE THE FINAL PAYABLE AMOUNT.

FOR THE DETAILED RECEIPT, THE PROGRAM CHECKS EACH ITEM COUNTER (I1-I10). IF A QUANTITY IS GREATER THAN ZERO, IT PRINTS THAT ITEM'S NAME, QUANTITY, AND TOTAL PRICE. FINALLY, THE PROGRAM PRINTS A STRUCTURED BILL SUMMARY SHOWING THE ORIGINAL BILL, DISCOUNT APPLIED, AMOUNT AFTER DISCOUNT, SERVICE CHARGE, GST, AND THE FINAL PAYABLE AMOUNT.

INPUT FROM THE USER

PROCESSING LOGIC

OUTPUT GENERATING

ALGORITHM AND CODE

Step 1: Start the Program
Step 2: Declare Variables: bill, choice , quantity, couponChoicE
Step 3: Display Menu
Step 4: Take User Order
Step 5: Calculate Bill Using Switch Case
Step 6: Ask to Continue
Step 7: Display Total Bill Before Discount
Step 8: Display Coupon Menu
Step 9: Take Coupon Choice
Step 10: Calculate Discount
Step 11: Calculate Amount After Discount
Step 12: Calculate Service Charge (10%)
Step 13: Calculate GST (5%)
Step 14: Calculate Final Amount
Step 15: Print Detailed Bill
Step 16: Stop

```c
#include <stdio.h>

int main() {

    int bill = 0;
    int choice, quantity;
    int continueOrder;
    int couponChoice;
    float discountAmount = 0;
    int i1,i2,i3,i4,i5,i6,i7,i8,i9,i10;
    i1=i2=i3=i4=i5=i6=i7=i8=i9=i10=0;

    do {

        printf("\n------ MENU ------\n");
        printf("1. Pizza      - Rs.150\n");
        printf("2. Burger     - Rs.90\n");
        printf("3. Pasta      - Rs.120\n");
        printf("4. Sandwich   - Rs.80\n");
        printf("5. Fries      - Rs.90\n");
        printf("6. Vada Pav   - Rs.20\n");
        printf("7. Tacos      - Rs.75\n");
        printf("8. Salad      - Rs.90\n");
        printf("9. Samosa     - Rs.25\n");
        printf("10. Wrap      - Rs.115\n");
        printf("-------------------\n");

        printf("Enter your choice (1-10): ");
        scanf("%d", &choice);
```

```c
printf("Enter quantity: ");
scanf("%d", &quantity);

switch(choice) {
    case 1:bill += 150 * quantity;
           i1 += quantity;
           break;
    case 2:bill += 90 * quantity;
           i2 += quantity;
           break;
    case 3: bill += 120 * quantity;
            i3 += quantity;
            break;
    case 4: bill += 80 * quantity;
            i4 += quantity;
            break;
    case 5:bill += 90 * quantity;
           i5 += quantity;
           break;
    case 6:bill += 20 * quantity;
           i6 += quantity;
           break;
    case 7: bill += 75 * quantity;
            i7 += quantity;
            break;
    case 8: bill += 90 * quantity;
            i8 += quantity;
            break;
    case 9:bill += 25 * quantity;
           i9 += quantity;
```

```c
main.c
        case 10:bill += 115 * quantity;
                i10 += quantity;
                break;
        default:
            printf("Invalid choice!\n");
    }

    printf("Do you want to order anything else? (1 = Yes, 0
        = No): ");
    scanf("%d", &continueOrder);

} while(continueOrder == 1);

printf("\nTotal Bill Before Discount = Rs.%d\n", bill);

// Coupon System Using int
printf("\n------ COUPON MENU ------\n");
printf("1. SAVE10  (10%% Discount)\n");
printf("2. SAVE20  (20%% Discount)\n");
printf("3. FLAT50  (Rs.50 Discount)\n");
printf("4. No Coupon\n");
printf("--------------------------\n");

printf("Enter coupon choice (1-4): ");
scanf("%d", &couponChoice);

switch(couponChoice) {

    case 1:
        discountAmount = bill * 0.10;
```

```c
    case 1:discountAmount = bill * 0.10;
            break;
    case 2:discountAmount = bill * 0.20;
            break;
    case 3:discountAmount = 50;
            break;
    case 4:discountAmount = 0;
            break;
    default:
        printf("Invalid coupon! No discount applied.\n");
        discountAmount = 0;
}

float afterDiscount = bill - discountAmount;
float serviceCharge = afterDiscount * 0.10;
float gst = (afterDiscount + serviceCharge) * 0.05;
float finalAmount = afterDiscount + serviceCharge + gst;
printf("\nBill after adding 10%% service charges & 5%% GST:
    \n");

printf("\n------- BILL DETAILS --------\n");
printf(" Item      qty      price\n");
if (i1 > 0){
    printf(" Pizza      %d      %d\n", i1, i1*150);
}
if (i2 > 0){
    printf(" Burger     %d      %d\n", i2, i2*90);
}
if (i3 > 0){
    printf(" Pasta      %d      %d\n", i3, i3*120);
```

```c
}
if (i4 > 0){
    printf(" Sandwich   %d      %d\n", i4, i4*80);
}
if (i5 > 0){
    printf(" Fries      %d      %d\n", i5, i5*90);
}
if (i6 > 0){
    printf(" Vada Pav   %d      %d\n", i6, i6*20);
}
if (i7 > 0){
    printf(" Taco       %d      %d\n", i7, i7*75);
}
if (i8 > 0){
    printf(" Salad      %d      %d\n", i8, i8*90);
}
if (i9 > 0){
    printf(" Samosa     %d      %d\n", i9, i9*25);
}
if (i10 > 0){
    printf(" Wrap       %d      %d\n", i10, i10*115);
}
printf("\n------------------------------\n");

printf("Original Bill        : Rs.%d\n", bill);
printf("Discount             : Rs.%.2f\n", discountAmount);
printf("After Discount       : Rs.%.2f\n", afterDiscount);
printf("Service Charge (10%%) : Rs.%.2f\n", serviceCharge);
printf("GST (5%%)             : Rs.%.2f\n", gst);
printf("------------------------------\n");
```

```c
    printf("Final Payable Amount : Rs.%.2f\n", finalAmount);

    return 0;
}
```

Output

```
------ MENU ------
1. Pizza      - Rs.150
2. Burger     - Rs.90
3. Pasta      - Rs.120
4. Sandwich   - Rs.80
5. Fries      - Rs.90
6. Vada Pav   - Rs.20
7. Tacos      - Rs.75
8. Salad      - Rs.90
9. Samosa     - Rs.25
10. Wrap      - Rs.115
-------------------
Enter your choice (1-10): 7
Enter quantity: 7
Do you want to order anything else? (1 = Yes, 0 = No): 1
```

```
------ MENU ------
1. Pizza      - Rs.150
2. Burger     - Rs.90
3. Pasta      - Rs.120
4. Sandwich   - Rs.80
5. Fries      - Rs.90
6. Vada Pav   - Rs.20
7. Tacos      - Rs.75
8. Salad      - Rs.90
9. Samosa     - Rs.25
10. Wrap      - Rs.115
-------------------
Enter your choice (1-10): 1
Enter quantity: 8
Do you want to order anything else? (1 = Yes, 0 = No): 1
```

```
------ MENU ------
1. Pizza      - Rs.150
2. Burger     - Rs.90
3. Pasta      - Rs.120
4. Sandwich   - Rs.80
5. Fries      - Rs.90
6. Vada Pav   - Rs.20
7. Tacos      - Rs.75
8. Salad      - Rs.90
9. Samosa     - Rs.25
10. Wrap      - Rs.115
-------------------
Enter your choice (1-10): 5
Enter quantity: 1
Do you want to order anything else? (1 = Yes, 0 = No): 1
```

```
------ MENU ------
1. Pizza      - Rs.150
2. Burger     - Rs.90
3. Pasta      - Rs.120
4. Sandwich   - Rs.80
5. Fries      - Rs.90
6. Vada Pav   - Rs.20
7. Tacos      - Rs.75
8. Salad      - Rs.90
9. Samosa     - Rs.25
10. Wrap      - Rs.115
-------------------
Enter your choice (1-10): 10
Enter quantity: 2
Do you want to order anything else? (1 = Yes, 0 = No): 1
```

```
------ MENU ------
1. Pizza      - Rs.150
2. Burger     - Rs.90
3. Pasta      - Rs.120
4. Sandwich   - Rs.80
5. Fries      - Rs.90
6. Vada Pav   - Rs.20
7. Tacos      - Rs.75
8. Salad      - Rs.90
9. Samosa     - Rs.25
10. Wrap      - Rs.115
-------------------
Enter your choice (1-10): 8
Enter quantity: 5
Do you want to order anything else? (1 = Yes, 0 = No): 0
```

Case 1: Invalid Coupon Choice Entered

```
Total Bill Before Discount = Rs.2495

------ COUPON MENU ------
1. SAVE10   (10% Discount)
2. SAVE20   (20% Discount)
3. FLAT50   (Rs.50 Discount)
4. No Coupon
---------------------------
Enter coupon choice (1-4): 5
Invalid coupon! No discount applied.
```

```
Bill after adding 10% service charges & 5% GST:

------- BILL DETAILS --------
 Item      qty      price
 Pizza      8       1200
 Fries      1       90
 Taco       7       525
 Salad      5       450
 Wrap       2       230

-----------------------------
Original Bill         : Rs.2495
Discount              : Rs.0.00
After Discount        : Rs.2495.00
Service Charge (10%)  : Rs.249.50
GST (5%)              : Rs.137.23
-----------------------------
Final Payable Amount  : Rs.2881.73
```

Case 2: Valid Coupon Choice Entered

```
Total Bill Before Discount = Rs.2495

------ COUPON MENU ------
1. SAVE10   (10% Discount)
2. SAVE20   (20% Discount)
3. FLAT50   (Rs.50 Discount)
4. No Coupon
---------------------------
Enter coupon choice (1-4): 2
```

```
Bill after adding 10% service charges & 5% GST:

------- BILL DETAILS --------
 Item      qty      price
 Pizza      8       1200
 Fries      1       90
 Taco       7       525
 Salad      5       450
 Wrap       2       230

-----------------------------
Original Bill         : Rs.2495
Discount              : Rs.499.00
After Discount        : Rs.1996.00
Service Charge (10%)  : Rs.199.60
GST (5%)              : Rs.109.78
-----------------------------
Final Payable Amount  : Rs.2305.38
```

V&RO HOSPITALITY PVT LTD
PLAN B
PLAN B
No 942, 21st Main Rd, Siddanna
Layout,Banashankari
Bengaluru - 560070
GSTIN:29AAGCV4390M1ZL

Bill No: 2314PBSK/23-24
Stw: SUMIT Date: 30-05-23 Time:15.19
Table No: 44 Covers: 5

SNo	Description	Qty	Rate	Amount
1	GINGER ALE	3	130.00	390.00
2	Brownie	2	230.00	460.00
3	Seasoned Fries	1	270.00	270.00
4	Tuesday 1 Dozer	1	315.00	315.00
	NAKED PERI PERI			0.00
5	Tuesday 1/2 Doz	10	225.00	2250.00
	NAKED PERI PERI			0.00
	SPICY GARLIC			0.00
	FIRE CRACKER			0.00
	ABS			0.00
	FLAMING JALAPENO			0.00
	FIRE CRACKER			0.00
	CREAMY BUFFALO			0.00
	HONEY CHILLI			0.00
	ABS			0.00
6	MANGOBLOOM*	1	280.00	280.00
7	TEETOTTALERS	1	280.00	280.00
8	ELECTRIC CURF	1	280.00	280.00
	Total Amount			4525.00
	SERC @ 10%			452.50
	State Gst @ 2.5%			124.49
	Central Gst @ 2.5%			124.49
	Round Off			-0.48
	Net Amount			5226.00

KOT NO : 4797,4806,4834,4852
Total Items : 20
User ID: SHASHI

Thank You visit again

CONCLUSION

This program demonstrates a simple restaurant billing system developed using core C programming concepts such as loops, conditional statements, switch–case structures, and arithmetic operations. It allows users to place multiple orders, apply discounts through coupons, and generates a clear, itemized bill including service charges and GST. The program shows how structured programming can be used to create practical, real-world applications like automated billing systems.

FUTURE SCOPE/APPLICATIONS

- RESTAURANT & CAFÉ BILLING SYSTEMS – CAN BE EXPANDED INTO A REAL BILLING SOFTWARE FOR SMALL FOOD BUSINESSES AND LOCAL EATERIES.
- FOOD COURT / COLLEGE CANTEEN MANAGEMENT – USEFUL FOR HANDLING MULTIPLE ORDERS QUICKLY AND GENERATING AUTOMATED RECEIPTS.
- ONLINE FOOD ORDERING SYSTEMS – CAN SERVE AS THE BACKEND LOGIC FOR APPS OR WEBSITES THAT MANAGE MENUS, PRICING, AND BILLING.
- RETAIL SHOP BILLING SOFTWARE – WITH SMALL MODIFICATIONS, IT CAN BE ADAPTED FOR SUPERMARKETS OR SMALL RETAIL STORES.
- POS (POINT OF SALE) SYSTEMS – FORMS THE BASIC LOGIC FOR MODERN BILLING MACHINES USED IN RESTAURANTS AND STORES.

Thank You!